

HEURISTIC EVALUATION SUMMARY

Lumen Chart — EHR Clinical Documentation Redesign

Interactive Prototype (Clinician Flow) · Expert Heuristic Review · July 2026

Method	Frameworks	Scope	Date
Expert Heuristic Evaluation	Nielsen + PAIR + HAX + NIST + WCAG	Dashboard → Snapshot → SOAP → Pre-sign	July 10, 2026

1. Severity Summary

Findings were rated using Nielsen’s 0–4 severity scale.

Severity 0	Severity 1	Severity 2	Severity 3	Severity 4
No issue / Positive	Cosmetic only	Minor problem	Major problem	Usability catastrophe
14 findings	3 findings	7 findings	4 findings	0 findings

No catastrophic (Severity 4) issues were found. The prototype is safe for pilot testing with clinicians.

2. Top Findings (Highest Priority)

The following issues were rated Severity 3 (Major) and should be addressed before broader testing.

#	Heuristic	Finding	Impact
1	H3 · User Control & Freedom	No Unaccept / Re-open control after a SOAP section is Accepted	Creates false sense of completion and undermines trust for high-skepticism clinicians
2	H5 · Error Prevention	Pre-sign acknowledgments can be raced through with pure checkbox clicks	Weakens the two-gate safety model intended to prevent signing of unreviewed AI content
3	H9 · Error Recovery	Weak failure + retry path for “Writing to chart...” and gate failures	Users can become stuck with no clear recovery if a network or state error occurs
4	H3 · User Control	No clear Cancel visit / Discard ambient draft path mid-flow	Clinicians have limited freedom if they decide the capture is unusable

3. Key Strengths

- Amber AI provenance + per-section Accept gates + two-gate signing is an exemplary implementation of AI transparency & trust calibration (directly addresses Evidence Wall D1–D5).

- Strong visual hierarchy for critical alerts (queue-level surfacing) reduces classic alert-fatigue risk.
- Clinical language and guideline-sourced “Why?” tooltips create excellent match to the real world.
- Real-time feedback (badge flips, toasts, disabled Sign button) provides good visibility of system status.
- Simplified view toggle and Accept All accelerator support both novice and expert efficiency.

4. Priority Recommendations

P0 — Fix before pilot

- Add Unaccept / Re-open for review control on each SOAP section (and global option).
- Strengthen the pre-sign gate so pure checkbox-racing is harder (micro-scroll, hold-to-confirm, or progressive disclosure).
- Add explicit failure + retry UI for the “Writing to chart...” state and any failed gate.

P1 — High value for learning

- Persist a Visit Context sticky (checked agenda + open SDOH + patient header) while in SOAP and Pre-sign.
- Show exact remaining sections (“2 of 4 still need Accept”) in the disabled Sign button or progress indicator.
- Add keyboard accelerators (Accept next, toggle transcript, jump to next unreviewed).
- Expose a lightweight “How this AI works / Trust model” drawer.

5. Evaluation Method (Brief)

This summary is based on a full expert heuristic evaluation using:

- Nielsen’s 10 Usability Heuristics (2020 refined explanations)
- Google PAIR Guidebook + Microsoft HAX Toolkit (AI-specific transparency, control, and trust)
- WCAG 2.2 AA (accessibility)
- NIST EHR usability guidance (clinical safety)
- Project Evidence Wall design recommendations (D1–D5)

— End of Summary —

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